COMSATS University Islamabad
Sahiwal Campus
(Department of Software Engineering)

Course Title:	Object Oriented Programming Lab			**Course Code:**	CSC241	**Credit Hours:**	4
Course Instructor:	Mr. Shehzad Ali			**Program Name:**	BSSE		
Semester:	3rd	**Batch:** FA-24	**Section:**	B	**Date:**		
Time Allowed:				**Maximum Marks:**	10		
Student's Name:				**Reg. No.**	CUI/ /SWL		

Important Instructions / Guidelines:

Read the question paper carefully and answer the questions according to their statements.
Mobile phones are not allowed. Calculators must not have any data/equations etc. in their memory.

Assignment no.3 Fall-2025

Question No. 1 **Marks (5) (CLO-4, C3, GA 2-4)**

Develop a program that demonstrates Object-Oriented Programming (OOP) concepts by creating a system for managing an online store. The program should include the following features:

1. **Define a class Product** with attributes such as name, price, category, and stock_quantity. Implement methods to display product details and check if the product is in stock.
2. **Create a subclass Electronics** that extends Product. Add additional attributes such as warranty_period and brand. Override the display_product_info() method to display specific details for electronics.
3. **Create another subclass Clothing** that extends Product. Add attributes such as size and color. Override the display_product_info() method to display specific details for clothing.
4. **Implement an interface Discountable** that includes a method apply_discount() which calculates the discount for a product. Implement this interface in both Electronics and Clothing classes, where each class applies a different discount logic (e.g., 10% for Electronics and 20% for Clothing).
5. **Implement a class ShoppingCart** that can hold multiple Product objects. Include methods to add products, remove products, and calculate the total price of the cart, considering the discount if the product implements the Discountable interface.
6. **Use polymorphism** to demonstrate how the display_product_info() method is overridden in different subclasses of Product, and how the correct method is called based on the product type.
7. **Encapsulate the attributes** in each class and provide getter and setter methods to safely access or modify them.

Expected Outcome:

The program should reflect how OOP principles like inheritance, polymorphism, encapsulation, and interfaces are applied to a real-world scenario such as an online store. The system should be modular, maintainable, and easily extendable, demonstrating how interfaces can be used to enforce a contract for specific behaviors (e.g., applying discounts) across different classes.